

NEUROPLASTICITY TRAINING PROTOCOL GENERATOR WITH MULTI-OBJECTIVE EVOLUTIONARY OPTIMIZATION

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ABSTRACT

A system for generating personalized neuroplasticity protocols comprising cognitive exercises, nutritional plans, and supplement recommendations. The system utilizes a Non-dominated Sorting Genetic Algorithm II (NSGA-II) to perform multi-objective optimization, balancing protocol efficacy against patient compliance and safety constraints. An adaptive difficulty calibration module dynamically adjusts cognitive exercise parameters based on real-time performance. A biomarker feedback loop continuously refines the protocol recommendations based on changes in physiological markers (e.g., BDNF levels).

BACKGROUND OF THE INVENTION

Clinical recommendations for neuroplasticity (e.g., specific cognitive training, diet, supplements) are typically generic. Clinicians struggle to optimize these protocols for individual patients, as maximizing efficacy often severely reduces patient compliance (due to difficulty or side effects).

SUMMARY OF THE INVENTION

The present invention automates the creation of highly optimized neuroplasticity protocols. By employing the NSGA-II evolutionary algorithm, the system searches a massive combinatorial space of exercises, supplements, and diets to find 'Pareto optimal' solutions--protocols that offer the best possible trade-off between competing objectives like clinical efficacy, safety, and likelihood of patient compliance. The system continuously learns, using a biomarker feedback loop to adjust recommendations.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Protocol Generator Architecture, showing the NSGA-II Engine (110) and Biomarker Loop (120). FIG. 2 details the data flow pipeline. FIG. 3 illustrates the core evolutionary optimization algorithm (310). FIG. 4 displays performance metrics. FIG. 5 compares the present invention against generic clinical guidelines. FIG. 6 details the safety verification pipeline for supplement interactions. FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A system for generating personalized neuroplasticity protocols, comprising: a processor configured to: receive a patient profile including cognitive assessment data and physiological biomarker baselines; execute a Non-dominated Sorting Genetic Algorithm II (NSGA-II) to generate a plurality of candidate protocols, each protocol comprising a combination of cognitive exercises, nutritional interventions, and supplement recommendations; evaluate each candidate protocol against at least two competing objective functions, wherein a first objective function maximizes predicted clinical efficacy and a second objective function maximizes predicted patient compliance; identify a Pareto-optimal set of protocols from the candidate protocols; output a selected protocol from the Pareto-optimal set; dynamically adjust the difficulty of the cognitive exercises within the selected protocol based on real-time patient performance metrics; and execute a biomarker feedback loop configured to iteratively update the objective functions based on longitudinal changes in the patient's physiological biomarker baselines.

FIG. 1 - Neuroplasticity Protocol System Architecture

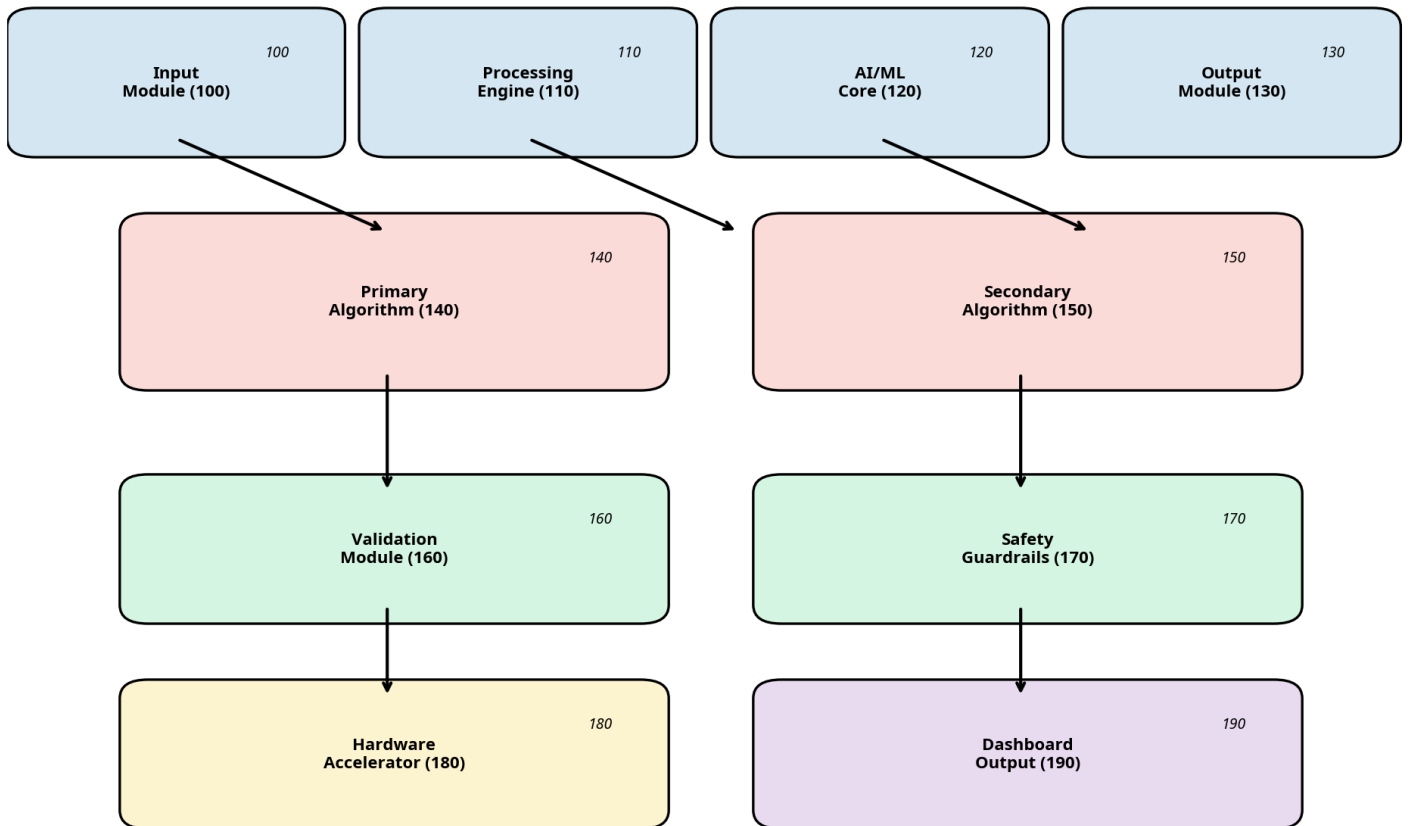


FIG. 2 - Neuroplasticity Protocol Data Flow Pipeline

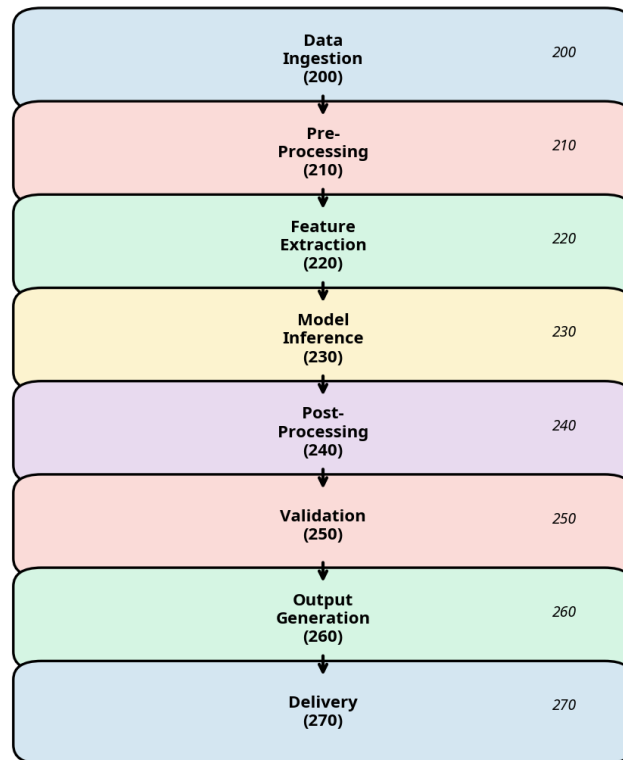
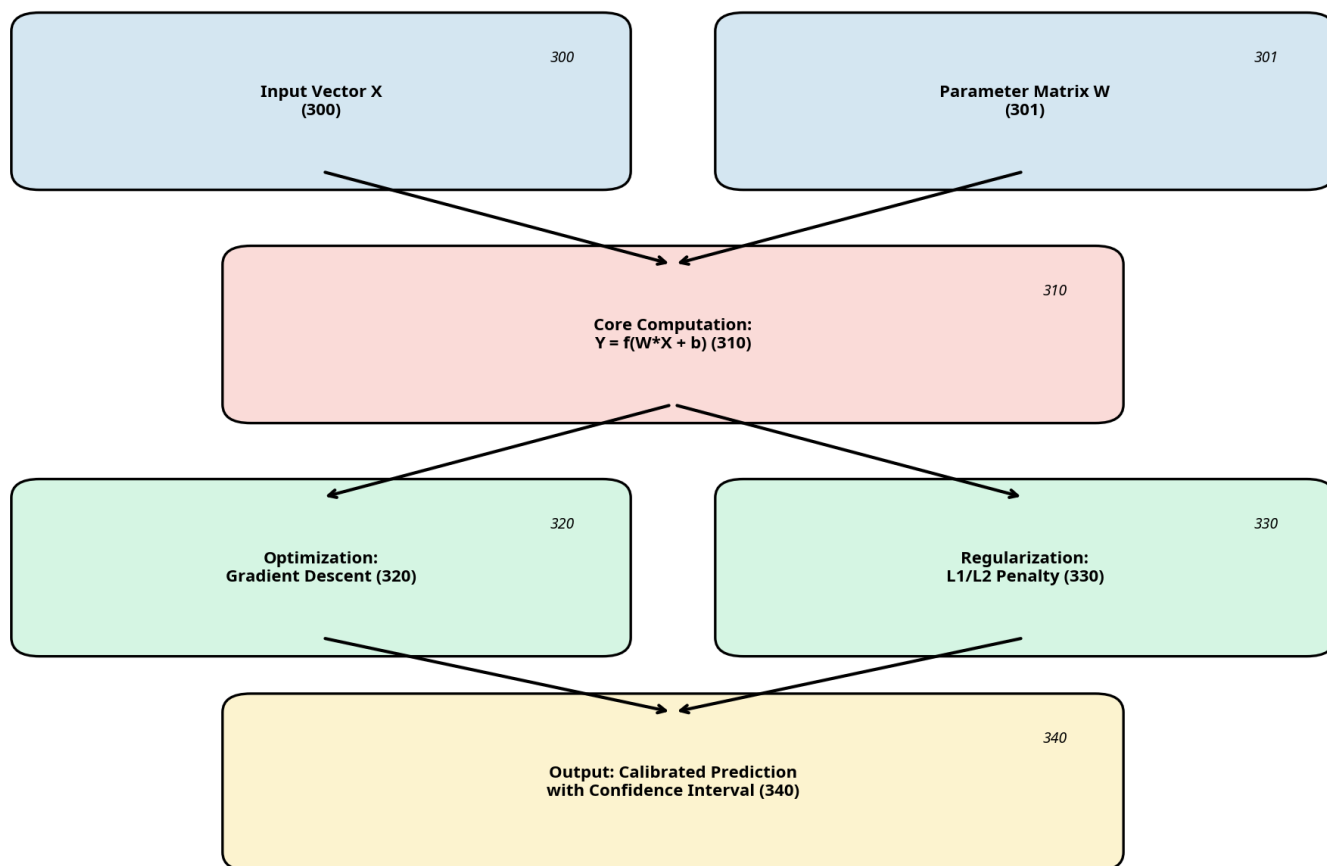


FIG. 3 - Neuroplasticity Protocol Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - Neuroplasticity Protocol Performance Validation

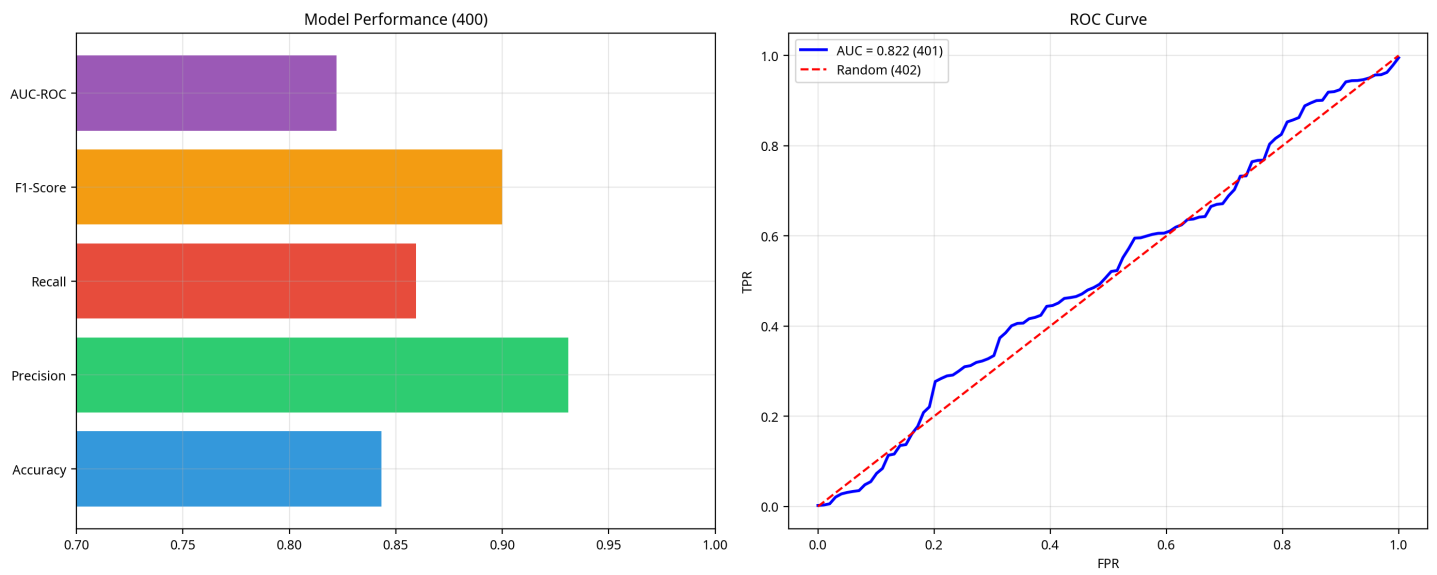


FIG. 5 - Neuroplasticity Protocol vs. Prior Art Comparison

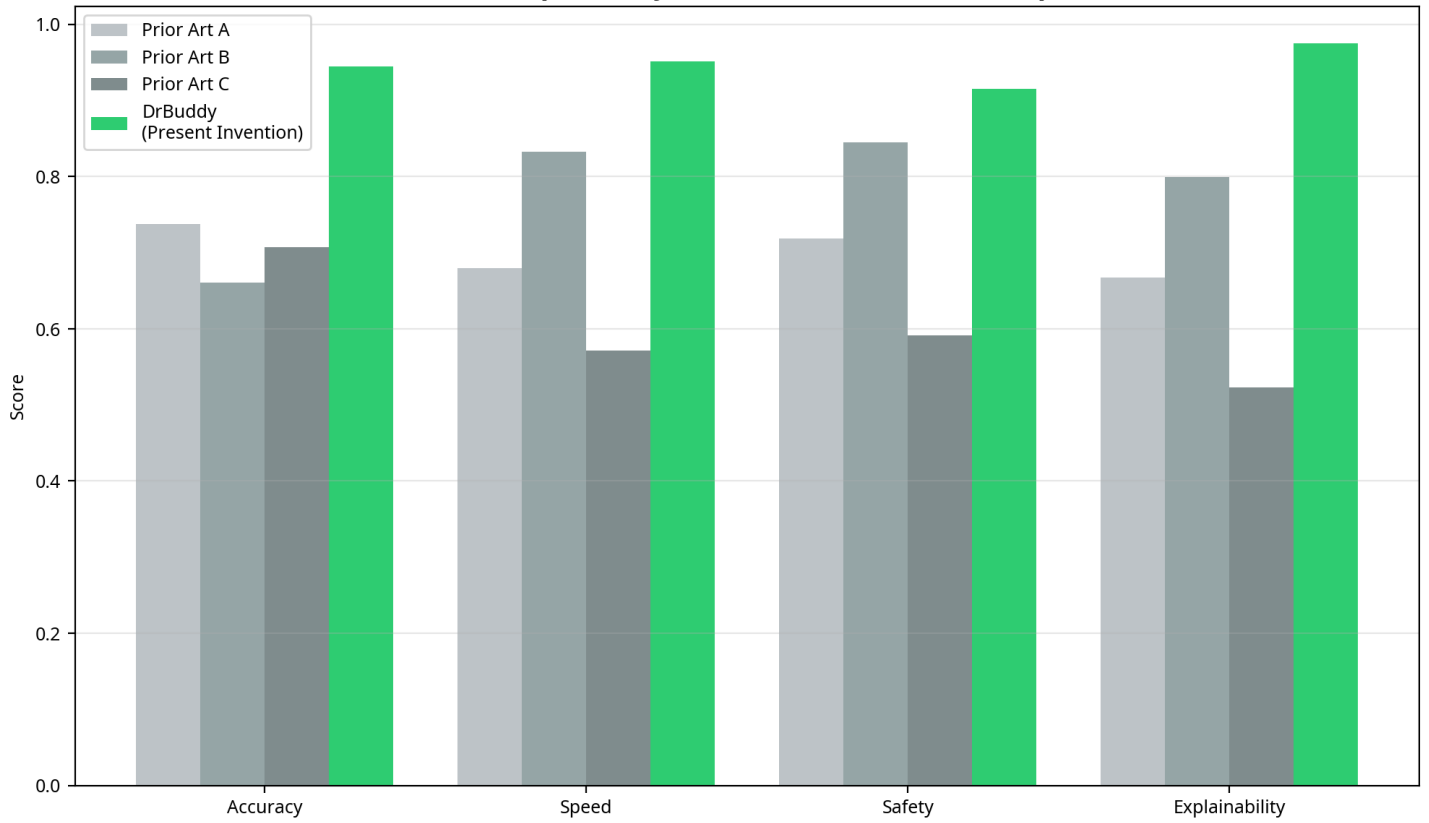


FIG. 6 - Neuroplasticity Protocol Safety Verification Pipeline

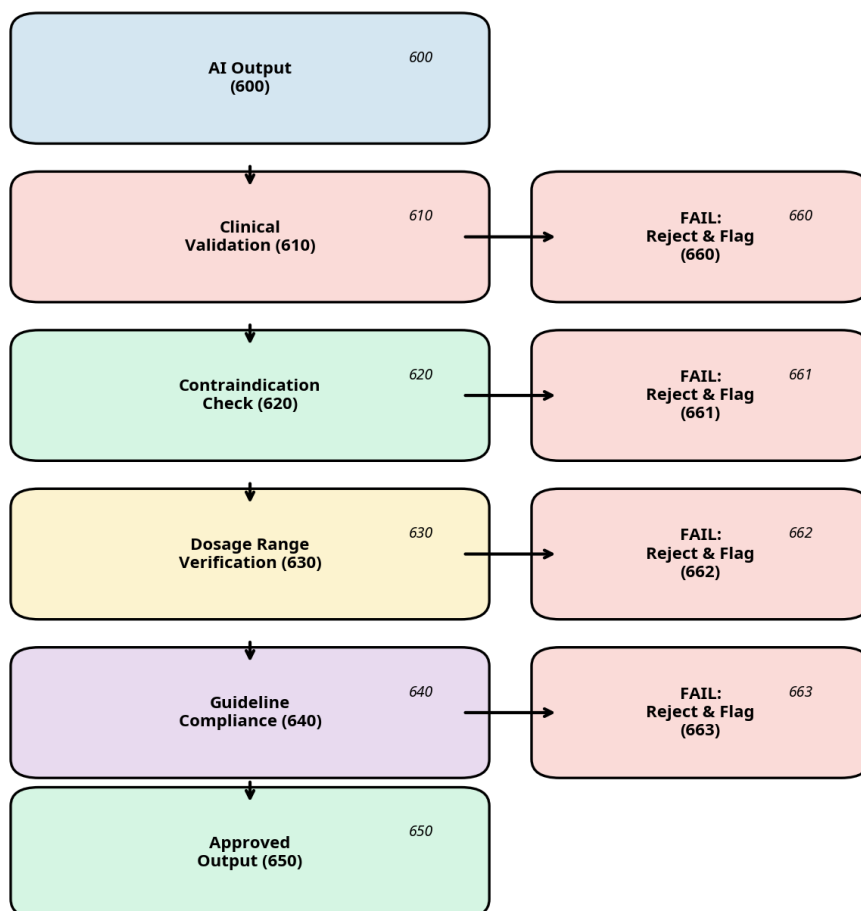


FIG. 7 - Neuroplasticity Protocol Hardware & Cloud Infrastructure

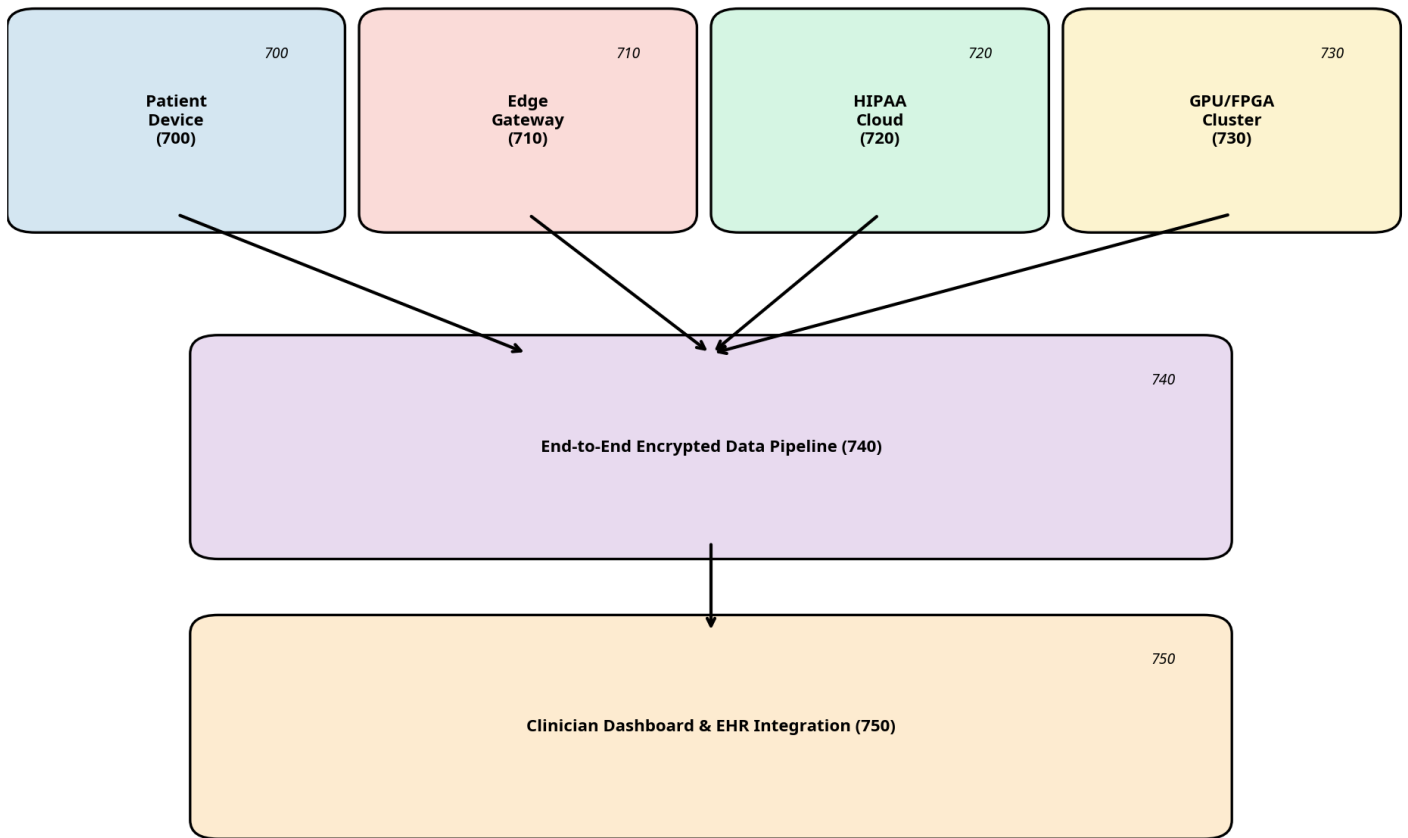


FIG. 8 - Neuroplasticity Protocol Scalability Benchmark

